Lecture 11

The Biological Database Landscape

From raw data to biological knowledge

Computing for Data-Driven Biology · 5BI00A
Master's Programme in Bioinformatics · Umeå University

Session Aims & ILO Links

1	Describe the four-tier database hierarchy and explain how data becomes biological knowledge	ILO 5
2	Identify primary archives, reference databases, functional databases, and specialist resources	ILO 5
3	Distinguish GenBank/RefSeq and Swiss-Prot/TrEMBL; explain evidence codes in practice	ILO 5
4	Retrieve and parse sequences from NCBI and UniProt using the command line	ILO 1, 7
5	Commit data and documented commands to Git with meaningful commit messages	ILO 12

This session follows the HPC2N Linux/CLI block — the exercise applies and reinforces those skills.

Opening Poll

Which of these databases have you heard of?

NCBI	ENA	UniProt	Ensembl
GEO	InterPro	KEGG	STRING
AlphaFold	gnomAD	GBIF	None of the above

Also: In one word — what IS a biological database? (open word cloud)

Part 1

How does data become biological knowledge?

The Four-Tier Database Hierarchy

Tier 1 — Primary Archives

e.g. ENA · SRA · GenBank · GEO · PRIDE · MetaboLights

Raw experimental data deposited by researchers. Submissions-based, high-volume, minimal curation.

Tier 2 — Reference Databases

e.g. RefSeq · UniProt/Swiss-Prot · Ensembl · UCSC Genome Browser

Curated, versioned, annotated collections. Manual or automated curation. Higher confidence than Tier 1.

Tier 3 — Functional & Interpretive

e.g. GO · InterPro · KEGG · Reactome · STRING · AlphaFold DB · PDB

Add biological meaning: functional classification, pathway membership, interaction networks, domain architecture.

Tier 4 — Specialist / Organism-Specific

e.g. FlyBase · TAIR · PlantGenIE · gnomAD · ClinVar · SILVA · GBIF

Integrate all tiers with community-curated knowledge for a specific organism, disease, or domain.

Errors in lower tiers propagate upward. Always check the evidence basis of an annotation, regardless of which tier the database belongs to.

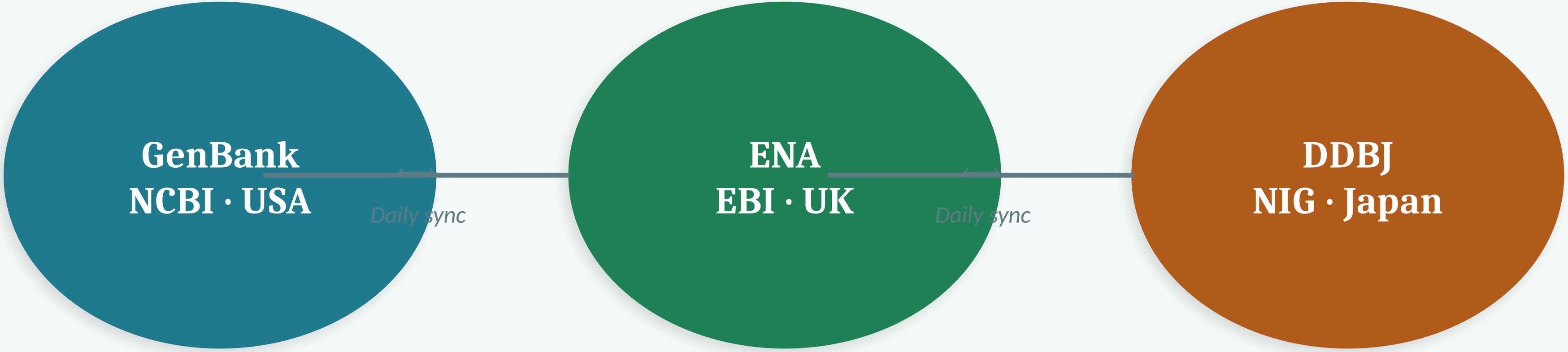
💬 *Pair Discussion — 4 minutes*

Think of a biological question you genuinely care about.

1. What kind of data would you need to answer it?
2. Where do you think that data might exist?
3. What format would it be in?
4. Would you expect it to be publicly available?

Brief share-back: one example per pair.

The Global Archiving Infrastructure: INSDC



🔄 All three databases synchronise daily — the same accession is available from any of them

📍 ENA is typically faster from European networks; NCBI tools integrate tightly with GenBank

🆔 Accession numbers are shared across INSDC partners — you do not need to know which database to search

Primary Archives: Sequence Data

Tier 1

GenBank

All publicly available nucleotide sequences. Submissions-based — any researcher can deposit. No filtering for quality or annotation completeness.

Accession format: "AY123456" (traditional) or "NM_000546" (RefSeq — see below). Used for: searching all known sequences of a gene across all species; BLAST searches.

Tier 2

RefSeq

NCBI's curated, non-redundant reference sequences. Manually reviewed and annotated by NCBI staff. Stable, versioned accessions.

Prefixes: NM_ (mRNA), NR_ (ncRNA), NP_ (protein), NC_ (chromosome). Use for: genome alignment, annotation, comparative genomics. Preferred reference for most analyses.

Tier 1

ENA / NCBI SRA

Raw sequencing reads — FASTQ files direct from sequencing instruments. The SRA is the most important archive for full reproducibility of NGS studies.

Accession formats: SRR/ERR/DRR (runs), SRS/ERS/DRS (samples), SRP/ERP/DRP (studies). If a paper deposited raw reads, you can reproduce the entire analysis.

Primary Archives: Expression, Proteomics, Metabolomics

GEO

Gene Expression Omnibus · NCBI

Microarray and RNA-seq expression data. Contains both raw data (FASTQ, linked to SRA) and processed data (count matrices, fold-changes). Key resource for FAIR assessment — metadata quality varies enormously.

ArrayExpress

EBI · European counterpart to GEO

Broad overlap with GEO; many datasets are cross-deposited. Part of the EMBL-EBI data ecosystem. Often the submission portal of choice for ERC-funded European projects.

PRIDE

PRoteomics IDEntifications Database · EBI

Primary archive for mass spectrometry-based proteomics raw data and results. Part of the ProteomeXchange consortium. If a paper presents proteomic data, its raw files should be here.

MetaboLights

Metabolomics archive · EBI

Open archive for metabolomics studies — NMR, LC-MS, GC-MS. Covers both raw spectral data and processed metabolite tables. Growing in importance with multi-omics integration.

GEO and ArrayExpress are the key resources for the FAIR metadata assessment exercise in Lecture 15.

Reference Databases: Ensembl and UniProt

Ensembl

ensembl.org · Tier 2

- Reference genome annotation portal by EBI/WTSI
- Gene models, regulatory annotation, comparative genomics
- Strong vertebrate coverage; Ensembl Genomes extends to plants, fungi, protists
- Standard reference for many bioinformatics pipelines
- Biomart tool: powerful query interface for large-scale data retrieval
- UCSC Genome Browser: similar scope, complementary tracks; particularly strong for human regulatory data (ENCODE, GTEx)

UniProt

uniprot.org · Tier 2

Swiss-Prot (reviewed)

~500K entries. Manually curated by expert annotators. High-confidence functional information. Always preferred when confidence matters.

TrEMBL (unreviewed)

~250M entries. Automatically annotated by rule-based pipelines. Valuable for coverage; treat annotations as predictions.

Evidence Codes: Reading the Fine Print

Almost all biological databases indicate HOW an annotation was derived. This is not a footnote — it is essential information for interpreting the annotation.

EXP / IDA / IMP / IPI / IGI — **Direct experimental evidence** — **High**

Biochemical assay, mutant phenotype, physical interaction, genetic interaction. Treat as established.

ISS / ISO / IBA — **Computational transfer via similarity** — **Moderate**

Annotation inferred from a homologous sequence in another organism. Accuracy depends on evolutionary distance and functional conservation.

IEA — **Inferred from Electronic Annotation** — **Lower**

Fully automated — no manual review. Derived by rule-based systems from sequence features or text mining. Correct more often than not, but must be stated as a prediction, not a fact.

⚠ **Before citing a functional annotation: check the evidence code. IEA and EXP are not equivalent.**

Quick Check — Before the Break

Q1. You have finished an RNA-seq experiment. Where do the raw FASTQ reads go?

A) GEO — Gene Expression Omnibus

B) NCBI SRA — Sequence Read Archive ✓

C) UniProt

D) Ensembl

Q2. A GO annotation is labelled IEA. What does this mean?

A) Inferred from Experimental Analysis — high confidence

B) Inferred from Electronic Annotation — automated, no manual review ✓

C) International Expert Assessment — specialist curator

D) Integrated Evidence Archive — aggregated from multiple sources

Q3. Which best describes the GenBank / RefSeq relationship?

A) They contain identical sequences with different names

B) RefSeq is curated and non-redundant; GenBank contains all submissions ✓

C) GenBank is European; RefSeq is American

D) They are entirely separate with no overlap

☕ Break — 10 minutes

Back at: ____________

After the break: functional databases · specialist resources · hands-on exercise

Part 2

Functional databases · Specialist resources · Hands-on

Functional & Interpretive Databases (Tier 3)

Gene Ontology (GO) *geneontology.org*	Controlled vocabulary for gene function across all species. Three aspects: Molecular Function, Biological Process, Cellular Component. Most GO annotations for non-model organisms are transferred computationally — check the evidence code.
InterPro *ebi.ac.uk/interpro*	Integrates Pfam, PRINTS, PROSITE, HAMAP, CDD, and others. Provides domain architecture for any protein sequence. Essential first step for characterising an unknown protein.
KEGG & Reactome *kegg.jp / reactome.org*	Pathway databases. KEGG: widely used, comprehensive, but restricts programmatic access. Reactome: open-source, manually curated, REST API available — preferred for reproducible analyses.
STRING *string-db.org*	Protein–protein interaction networks integrating co-expression, experimental, text-mining, and database evidence. Assign confidence scores. Widely used for contextualising gene lists from omics experiments.
AlphaFold DB & PDB *alphafold.ebi.ac.uk / rcsb.org*	PDB: experimentally determined 3D structures. AlphaFold DB: computational predictions for ~200M proteins (virtually all of UniProt). pLDDT score indicates per-residue confidence — essential to check before interpreting structural predictions.

Specialist Resources: Plant, Forest & Environment

JGI Phytozome *phytozome-next.jgi.doe.gov*	Primary portal for plant genome sequences and annotations. Standardised comparative genomics. Hosts Populus trichocarpa, Picea abies, Arabidopsis, and many others.
JGI MycoCosm *mycocosm.jgi.doe.gov*	JGI's equivalent for fungal genomes. Important for studies of mycorrhizal fungi, wood decay, and fungal ecology.
TAIR *arabidopsis.org*	The Arabidopsis Information Resource — gold standard for plant functional genomics. Source of much of the GO annotation propagated to other plant species.
PlantGenIE *plantgenie.se (in development)*	Specialist resource for conifer and angiosperm genomics developed at Umeå University. Integrates genome browsers, co-expression networks, and BLAST for boreal forest species including Picea abies and Populus tremula.
SILVA *arb-silva.de*	High-quality rRNA reference databases (16S/18S, 23S/28S). Primary reference for amplicon-based microbial community profiling and phylogenetic placement of environmental sequences.
UNITE *unite.ut.ee*	Fungal ITS sequence reference database. Provides species hypotheses for environmental fungal communities. Essential for soil ecology, mycorrhizal network studies, and biodiversity monitoring.
GBIF & BOLD *gbif.org / boldsystems.org*	GBIF: global biodiversity occurrence data — where organisms have been observed. BOLD: DNA barcode sequences linked to voucher specimens. Key for eDNA study design, macroecology, and SDG 14/15 monitoring.

Specialist Resources: Human, Clinical & Model Organisms

Human & Clinical

ENCODE *encodeproject.org*	Regulatory elements (TF binding, chromatin, histone modifications) across 100s of human cell types.
GTEx *gtexportal.org*	Tissue-specific expression and eQTLs across 54 human tissues. Reference for variant–expression relationships.
gnomAD *gnomad.broadinstitute.org*	Population variant frequencies across diverse human populations. Essential for assessing variant pathogenicity.
ClinVar *ncbi.nlm.nih.gov/clinvar*	Variant–phenotype relationships with clinical significance interpretations. Central for clinical variant classification.
OMIM *omim.org*	Manually curated database of human genes and genetic disorders. Detailed clinical and molecular descriptions.

Key Model Organism Databases

FlyBase *Drosophila melanogaster*	flybase.org
WormBase *Caenorhabditis elegans*	wormbase.org
ZFIN *Danio rerio (zebrafish)*	zfin.org
MGI *Mus musculus (mouse)*	informatics.jax.org
SGD *Saccharomyces cerevisiae*	yeastgenome.org
PomBase *Schizosaccharomyces pombe*	pombase.org

Model organism databases are the primary source of GO experimental annotations — they underpin functional inference across all species.

How to Evaluate Any Unfamiliar Database

You will encounter databases not covered here throughout your career. Apply this checklist.

1 Scope

What organism(s), data type(s), and biological level does it cover? Which tier of the hierarchy does it sit in?

2 Who maintains it?

Established institution (NCBI, EBI, JGI, model organism community) or a single research group? What is the funding source?

3 Update cycle & versioning

Continuously updated or versioned releases? For reproducibility, always record which version you used.

4 Evidence basis

How are annotations derived? Are evidence codes available? Can you trace an annotation back to its primary source?

5 Programmatic access

REST API? FTP? Command-line client? Essential for any analysis that scales beyond manual work.

6 How to cite it

Most databases have a primary citation paper. Not citing a database you use is poor scientific practice.

Hands-On Exercise

Programmatic database access on the command line

Exercise: Your First Programmatic Database Query

You will retrieve biological sequences from NCBI and UniProt using the command line, parse them with Linux tools, and commit your work to Git.

Part A

🖥️ Set up your workspace

Create a directory · Initialise Git · First commit

Part B

🧬 Fetch from NCBI Entrez

curl to fetch TP53 in FASTA and GenBank format · inspect with head, grep, wc

Part C

🔬 Fetch from UniProt REST

curl to fetch TP53 protein · parse annotation · count amino acids · check evidence codes

Part D

📝 Document and commit

Update README with findings and access date · git add · git commit with a meaningful message

Login to Kebnekaise via OnDemand before we begin. Full commands are in your handout.

Exercise Parts A & B — Setup and NCBI Entrez

Part A — Setup

```
$ mkdir -p ~/course/lecture11-databases && cd ~/course/lecture11-databases
$ git init
$ echo "# Lecture 11: Programmatic Database Access" > README.md
$ git add README.md && git commit -m "initial commit: lecture11 exercise setup"
```

Part B — Fetch TP53 from NCBI Entrez

```
# Fetch in FASTA format (NM_000546.6 = human TP53 RefSeq mRNA)
$ curl "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?
db=nucleotide&id=NM_000546.6&rettype=fasta&retmode=text" > TP53_mRNA.fasta
# Inspect with tools you already know
$ head -5 TP53_mRNA.fasta    # What is in the header?
$ grep -c ">" TP53_mRNA.fasta  # How many sequences?
$ wc -c TP53_mRNA.fasta       # File size in bytes?
# Fetch the same record in GenBank format (more metadata)
$ curl "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?
db=nucleotide&id=NM_000546.6&rettype=gb&retmode=text" > TP53_mRNA.gb
$ grep "DEFINITION\|SOURCE\|ORGANISM" TP53_mRNA.gb
```

Exercise Parts C & D — UniProt and Git Commit

Part C — Fetch TP53 protein from UniProt REST API

```
# P04637 = human TP53 in UniProt Swiss-Prot
$ curl "https://rest.uniprot.org/uniprotkb/P04637.fasta" > TP53_protein.fasta
$ head -3 TP53_protein.fasta                        # What does the UniProt header look like?
$ grep -v ">" TP53_protein.fasta | tr -d '\n' | wc -c  # How many amino acids?
# Fetch the full annotation record
$ curl "https://rest.uniprot.org/uniprotkb/P04637.txt" > TP53_protein.txt
$ grep "^ID" TP53_protein.txt                       # Is this Swiss-Prot or TrEMBL?
$ grep "^DR   GO;" TP53_protein.txt | head -8       # What GO terms are listed?
$ grep "^DR   GO;" TP53_protein.txt | grep -o "IDA\|IEA\|IMP\|EXP" | sort | uniq -c
```

Part D — Document and commit

```
# Update your README with findings, then:
$ git add .
$ git commit -m "lecture11: retrieved TP53 from NCBI and UniProt

  - NM_000546.6 fetched in FASTA and GenBank format via Entrez API
  - P04637 fetched from UniProt REST API
  - Documented access date and findings in README"
```

What You Should Know After This Session

- ✓ Describe the four-tier hierarchy and give one example from each tier
- ✓ Explain the INSDC and why the same nucleotide sequence appears in ENA, GenBank, and DDBJ
- ✓ Distinguish GenBank from RefSeq, and Swiss-Prot from TrEMBL
- ✓ Explain what an evidence code is and why IEA ≠ EXP in practice
- ✓ Name ≥ 2 databases from each of: primary archives · functional · plant/environmental · human/clinical
- ✓ Use curl to retrieve a sequence from NCBI Entrez and from UniProt, and explain the URL structure
- ✓ Use grep, wc, and tr to extract information from FASTA and GenBank format files
- ✓ Commit data and documented commands to Git with a meaningful commit message
- ✓ Apply the six-point checklist to evaluate any unfamiliar database